# CARLA PENAFIEL MESSMER

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## Education

**Columbia University** — New York, NY, USA
**M.S. in Data Science**; *DSI Scholar, Data Science Institute – "Mapping Children's Vulnerability"* — *Expected Dec. 2027*

**University of Chile** — Santiago, Chile
**M.A. in Economics**; *Class Rank: 2nd of 21, High Honors; Thesis: "Gender conditions, labor market and child abuse in Chile"* — 2023

**University of Chile** — Santiago, Chile
**B.S. in Industrial Engineering**; *Class Rank: 1st of 174, High Honors* — 2023

## Experience

**Professional Experience**

**Lead Data Scientist** — Dec 2024 - 2026
**Bain & Company** — *Santiago, Chile*

- Built an end-to-end marketing mix model for an Argentine wine company using Google Meridian, including client data collection, 100+ model iterations, validation, and a budget-optimization dashboard to assess channel performance and inform marketing spend.
- Led an end-to-end operational optimization workstream for a Caribbean oil client, analyzing large-scale data and delivering a Streamlit dashboard with data-processing workflows and automated alerts; deployed to the client and reused for similar oil projects at Bain.
- Developed a reusable price-elasticity methodology and standardized framework for pricing recommendations, now used across Bain projects and offices.

**Senior Data Scientist** — Aug 2022 - Dec 2024
**Bain & Company** — *Santiago, Chile*

- Built price-demand elasticity prediction models in Python to quantify pricing effects on demand and inform optimal pricing across industries.
- Developed fraud-detection and early-warning solutions in Python using LightGBM, NLP/BETO embeddings, anomaly detection, clustering, and supervised learning; built and deployed a Streamlit dashboard in client infrastructure to prioritize high-probability alerts.
- Built an insurance pricing optimization framework using LightGBM and Pyomo to maximize portfolio margin subject to renewal-rate constraints.

**Data Scientist** — Feb 2022 - Aug 2022
**Bain & Company** — *Santiago, Chile*

- Built gas pricing optimization models in R/RStudio with operational constraints, delivering a client-facing tool that generated recommended daily prices.
- Designed a competitor-price scraping and pricing optimization solution in Python for a large retailer, leaving the scraping and optimization pipeline with the client for ongoing use.

**Research Experience**

**Research Consultant** — 2021–2022
**Inter-American Development Bank (IDB)** — *Santiago, Chile*

- Conducted quantitative research on gender attitudes and perceptions among PhD holders in male-dominated fields, contributing to a co-authored study.

**Research Assistant** — 2020–2021
**J-PAL Latin America and the Caribbean (J-PAL LAC)** — *Santiago, Chile*

- Played a central role in the design and implementation of a randomized controlled trial on information frictions in mortgage refinancing; designed the customer statement (cartola) used across BancoEstado as part of the intervention and contributed to experimental data preparation and analysis.

## Skills

- **Programming:** Python (pandas, NumPy, scikit-learn, statsmodels, XGBoost, LightGBM), R, Stata, SQL
- **ML & Analytics:** NLP, agentic AI, anomaly detection, clustering, supervised learning, forecasting, predictive analytics, pricing optimization, Pyomo
- **Tools:** Git/GitHub, Power BI, Streamlit
- **Languages:** Spanish (native), Portuguese (advanced)

## Honors

- **Honor Student, FCFM, University of Chile** (2019–2022)
- **Women's Scholarship in Economics and Science** (2020–2022)
- **Honor Student, M.A. in Economics** (2023)

## Academic Activities

- **Teaching:** Graduate Workshop on Applied Economics, MAGCEA (2023–2026); Graduate Economics, MBA (2021–2024); Economics for Public Policy I, M.P.P., University of Chile (2026)
- **Publications:** Co-author, *Vida cotidiana en Pandemia por COVID-19 en Chile*, *Pandemia y Crisis* (2022); M.A. thesis manuscript in preparation, *Gender conditions, labor market and child abuse in Chile*